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Absolute dating of dedolomitization by means of paleomagnetic techniques

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ABSTRACT

Paleomagnetic analysis, in conjunction with petrographic studies, is an approach whereby absolute dates can be placed on diagenetic events. Hematite is associated with dedolomite in the Lower Ordovician Kindblade Formation in south-central Oklahoma and is interpreted as a by-product of the dedolomitization of ferroan dolomite. Paleomagnetic results date the time of dedolomitization as Late Pennsylvanian–Early Permian. Dedolomitization was probably caused by oxidizing fluids with high calcium contents that migrated through the Kindblade Formation following uplift of the Arbuckle Mountains.

INTRODUCTION

The value of diagenetic studies is commonly limited because the timing of events can be established only relatively. Paleomagnetic analysis, in conjunction with standard petrographic techniques, is an approach that in some cases can lead to the absolute dating of diagenetic events. If the magnetic direction carried by a ferromagnetic authigenic mineral such as hematite can be determined, the corresponding pole position can be compared to the apparent polar wandering path (APWP); hence, the time of remanence acquisition and presumably that of formation of the authigenic phase may be determined. In addition, if the authigenic magnetic phase can be texturally related to other diagenetic features, then the timing of those features can also be constrained. In this paper we report paleomagnetic and petrographic results that date the time of dedolomitization in the Lower Ordovician Kindblade Formation, south-central Oklahoma.

The Kindblade Formation is part of a thick sequence of shallow-water carbonate deposited in the southern Oklahoma aulacogen (Wickham, 1978) (Fig. 1). It consists of oolitic grainstones, intraclastic grainstones, lithoclastic grainstones, algal-sponge boundstones, fossiliferous wackestones, and burrowed mudstones deposited in subtidal to intertidal settings by both storm and fair-weather processes (Goldhammer and Elmore, 1983). The burrows in the mudstones are usually stained red by iron oxide minerals and contain abundant dedolomite.

Specimens of the burrowed mudstone were drilled using a portable coring apparatus and oriented in situ using a Brunton compass. Specimens were collected on the north and south flanks of the Arbuckle anticline in the Arbuckle Mountains, for use in a fold test (McElhinny, 1973). The results of the test would thereby constrain the time of remanence acquisition (and accordingly, that of dedolomitization) (Fig. 1). Specimens (2.2-cm length, 2.54-cm diam) were measured in a United Scientific three-axis cryogenic magnetometer. Thermal treatment was the most successful demagnetization technique and was applied to all specimens; specimens were heated using a Schonstedt TSD-1 unit in a stepwise manner (in 100 °C steps up to 300 °C and subsequently by 25 °C steps) until 5% or less of the magnetization remained. Results for each specimen were plotted in orthogonal projections and analyzed for vector-subtraction directions (Zijderveld, 1967). Isothermal-remanent-magnetization (IRM) acquisition and demagnetization experiments were also conducted to verify the magnetic mineralogy. Selective destructive demagnetization (SDD) (Larson, 1981), involving mechanical removal of the material filling the burrows, was also undertaken on some specimens.

Thin sections of samples were examined with both transmitted and reflected light. Samples were also studied by means of a scanning electron microscope that has energy-dispersive capabilities.

RESULTS

Petrographic analysis shows that the dedolomite occurs as composite calcite rhombohedra (Fig. 2A) having relict dolomite inclusions. Hematite, which is found as rosettes and plates less than 1 m in size, occurs in association with clay minerals surrounding the rhombohedra (Fig. 2B). Hematite and other iron oxides such as goethite are commonly as-

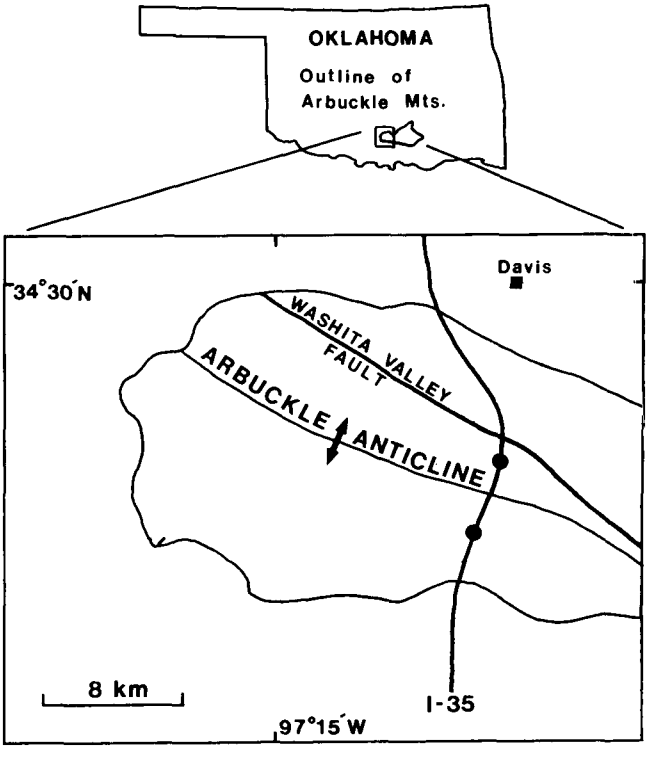


Figure 1. Generalized map of study area showing location of section on both flanks of Arbuckle anticline. Dots show collection locations.

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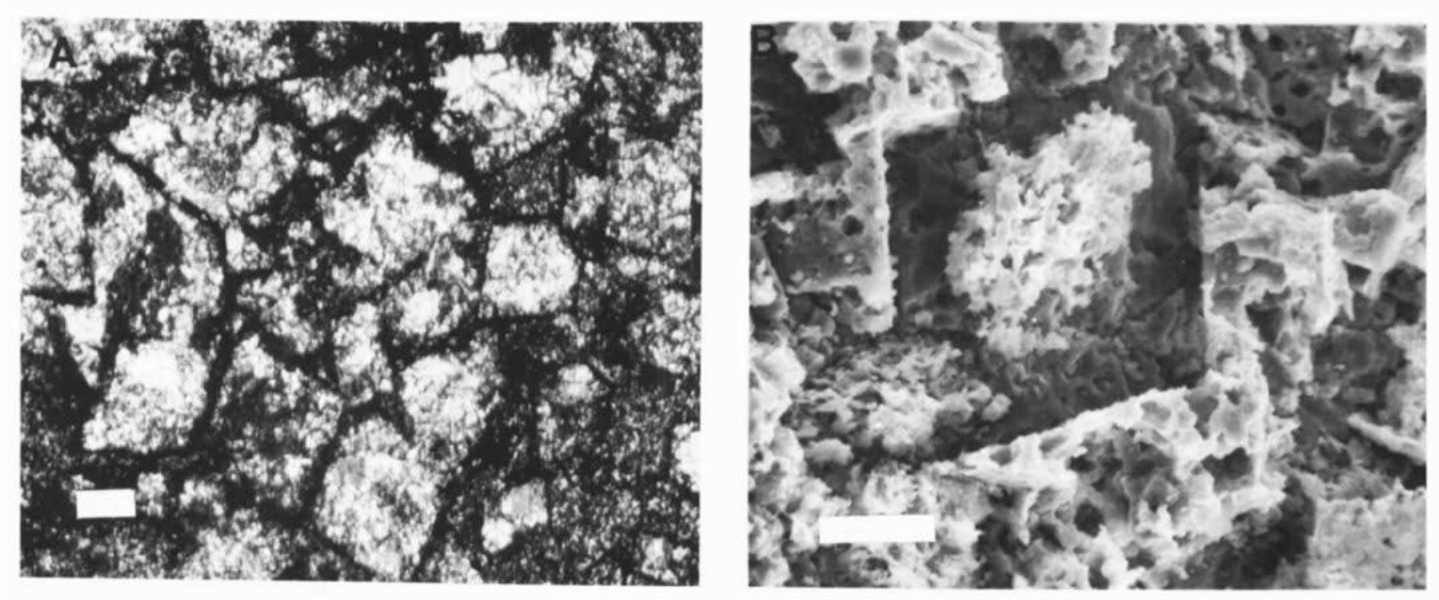


Figure 2. A: Composite calcite rhombohedra outlined by iron oxides and clay minerals. Bar scale = 100 μm. B: SEM photomicrograph of dedolomite crystal. Iron oxides and clays stand out in relief above calcite in etched sample. Bar scale = 100 μm.

sociated with dedolomite and are generally interpreted as by-products of dedolomitization of ferroan dolomite (e.g., Katz, 1971; Al-Hashimi and Hemingway, 1973; Frank, 1981).

The specimens exhibit varied behavior during thermal demagnetization (Figs. 3A, 3B). Most specimens contain only one remanence component, with a consistent southeasterly and shallow downward direction of magnetization before structural correction (Fig. 3A). Remanence directions for these specimens are plotted in Figure 4, and the statistics are listed in Table 1. For a few specimens (Fig. 3A) the magnetization increases at high temperatures and then a northwesterly and shallow-to-intermediate direction is removed, suggesting the presence of a second component.

Other specimens contain a direction that is easterly to southeasterly with moderate positive inclinations (Fig. 3B). During demagnetization the endpoints move toward a southeasterly and shallow direction, generally along a path moving away from the modern field direction. This behavior is illustrated on the Zijderveld diagram by a curved demagnetization path, which suggests simultaneous removal of two components. The east to southeasterly and intermediate downward direction is interpreted to be the vector sum of the southeasterly and shallow direction and a modern component. Specimens that exhibited this behavior were not used in the calculation of the statistics unless there was linear decay (at least three colinear points) of the southeasterly and shallow direction.

Remanence directions and site means prior to structural correction group well in a southeasterly and shallow direction (Fig. 4). After correction for the folding, the directions and site means diverge (Fig. 4). The fold test is positive and significant at the 99% level (McElhinny, 1964). The magnetization is therefore interpreted to have been acquired after the folding (Late Pennsylvanian). The pole corresponding to the mean direction (D = 149, I = 5 downward) is at 43°N, 128°E.

Maximum blocking temperatures in excess of 600 °C (Fig. 3A) indicate that hematite carries the magnetization. Isothermal remanent acquisition and subsequent demagnetization experiments confirm this conclusion. No evidence was found to indicate the presence of goethite or magnetite. In addition, removal of the hematite in the burrows using SDD resulted in decay of the southeasterly and shallow direction (Fig. 3C). This strongly suggests that the hematite associated with the dedolomite carries the magnetization.

DISCUSSION

The results described in the previous section indicate that the magnetization is secondary and resides in hematite associated with dedolomite. The mode of remanence acquisition, however, is a crucial and unresolved issue. Similar remanence directions in the West Spring Creek Formation, which directly overlies the Kindblade, were interpreted by Steiner (1973)

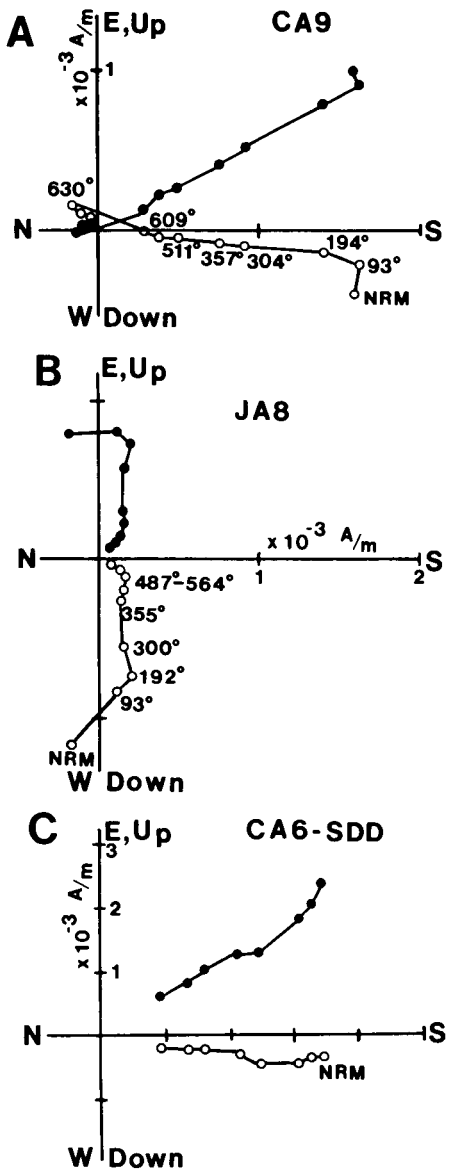


Figure 3. Demagnetization diagrams for Kindblade Formation. Closed circles represent projection on horizontal plane at various stages of demagnetization; open circles represent projections on vertical plane. A: Diagram showing decay of southeasterly and shallow direction. B: Diagram showing decay of magnetization interpreted as a vector sum of southeasterly direction and a modern component. C: Diagram illustrating decay of magnetization during SDD.

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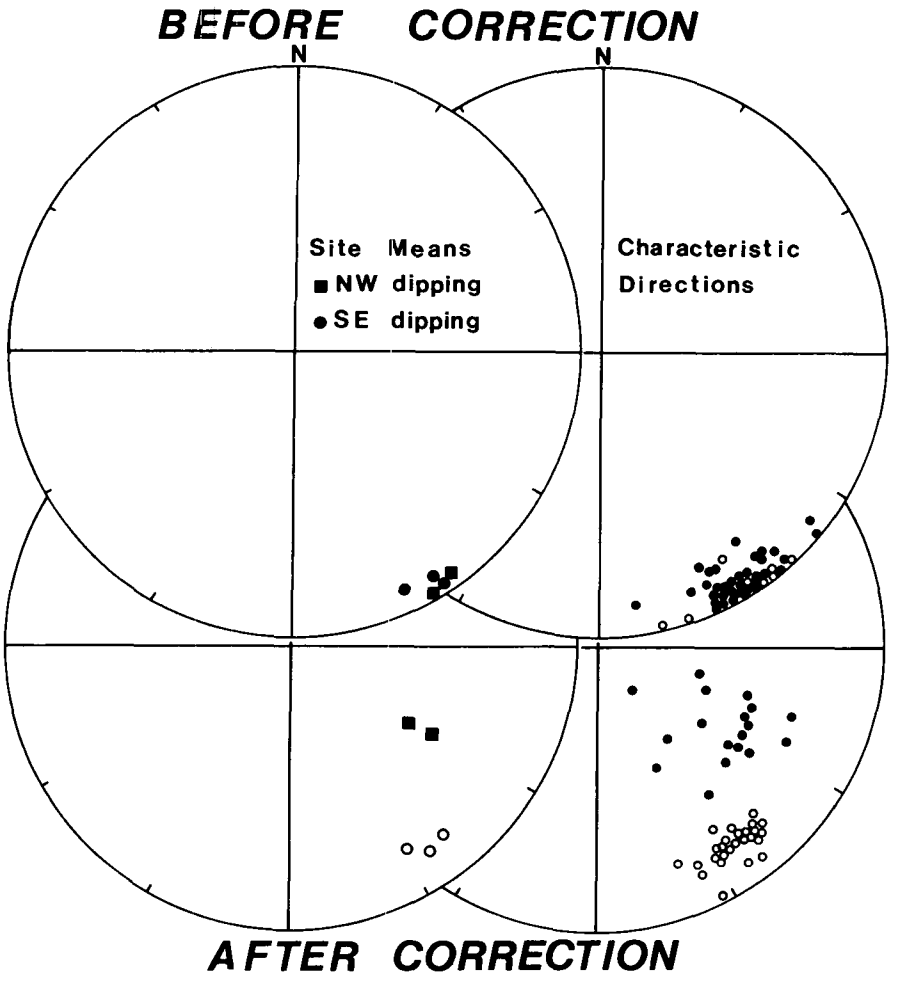


Figure 4. Equal-area projection of specimen directions and site means from Kindblade Formation for both before and after structural correction.

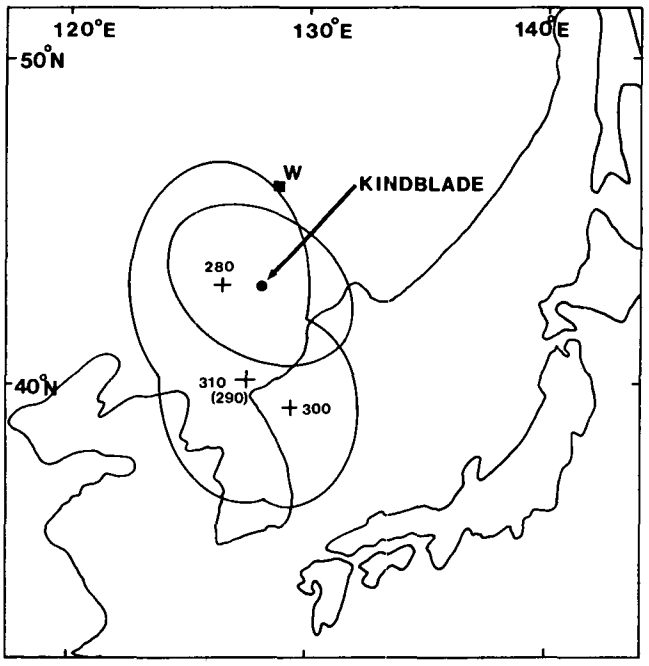


Figure 5. Kindblade paleopole with oval of 95% confidence (dp, dm) along with mean 280, 290, 300, 310 Ma paleopoles (30 Ma window) with ovals of confidence (Irving and Irving, 1982). Paleopole W is from West Spring Creek Formation (Steiner, 1973).

as a viscous partial thermal remanent magnetization (VpTRM) resulting from heating during deformation. The possibility of thermal remagnetization in both formations can be eliminated, however, in view of paleotemperature estimates based on conodonts. The conodont alteration index (CAI) for the Kindblade Formation in the Arbuckle anticline is 1.5 or less (R. L. Ethington, 1984, personal commun.). For any reasonable length of burial time (10^8 yr), maximum burial temperatures of less than 100 °C are indicated (Epstein et al., 1977). On the assumption of a burial temperature of 100 °C and a maximum burial time of 3×10^8 yr, the indicated maximum theoretical blocking temperature is about 400 °C (Pullaiah et al., 1975). Blocking temperatures in the Kindblade Formation extended to temperatures above 600 °C, thereby suggesting that the magnetization is not a VpTRM.

The magnetization is therefore interpreted to be a chemical remanent magnetization (CRM) residing in diagenetic hematite that was acquired during Late Pennsylvanian time or later; Ham (1978) suggested that the folding of the Arbuckle anticline is Virgilian (Late Pennsylvanian). The pole position (Fig. 5) is closely similar to North American paleopoles ranging from 280 to 310 Ma (Irving and Irving, 1982)—i.e., from Late Pennsylvanian to perhaps Early Permian.

The origin of the northwesterly component in some specimens is perplexing. Steiner (1973), in her study of the West Spring Creek Formation, noted that at high tempertures the remanence directions of some specimens moved toward the northwest. She tentatively interpreted the direction as the primary Ordovician magnetization. This interpretation could also apply to the Kindblade or, alternatively, the northwest direction may represent magnetization acquired during a reversed interval. Because so few specimens contain the high temperature component, the issue is difficult to resolve.

The diagenetic hematite in the Kindblade Formation probably precipitated directly when Fe^{2+} was released from ferroan dolomite during dedolomitization and oxidized to Fe^{3+}. Alternatively, the hematite may have dehydrated from a precursor ferric oxyhydroxide that formed during dedolomitization. If there was a precursor, then the paleomagnetic results would date the transformation to hematite, and not dedolomitization. This issue is difficult to evaluate because of uncertainties about the geochemistry and the stability of ferric oxyhydroxides. For example, there are few data on how long the dehydration reaction takes; there are also questions about whether particular reactions (e.g., goethite to hematite) occur in nature (Turner, 1980).In any case, the dehydration reaction is likely to take place readily in

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TABLE 1. SUMMARY OF PALEOMAGNETIC RESULTS

Site*	N/No†	Before tilt correction			After tilt correction			Pole position (before tilt)		dp§	dm§
		Decl/incl (°)	k	α95	Decl/incl (°)	k	α95				
Ca(S)	16/16	147/3	175	2.8	141/-15	175	2.8	43°N	131°E	1.4	2.8
Ga(S)	5/7	155/8	227	5.1	151/-18	227	5.1	45°N	120°E	2.6	5.1
HA(S)	5/5	148/8	122	6.9	145/-14	122	6.9	41°N	127°E	3.5	7.0
KZ(N)	10/12	148/2	31	8.8	124/49	31	8.8	44°N	130°E	4.4	8.8
KJMI(N)	8/18	145/4	31	10.2	122/40	31	10.2	41°N	132°E	5.1	10.2
Mean of sites	5/5	149/5	306	4.4	138/8	6	36.0	43°N	128°E	2.2	4.4
Mean of specimens	44/44	148/4	58	2.8	137/12	6	10.3	43°N	129°E	1.4	2.8

Note: Decl = declination, incl = inclination; k is Fisher's precision parameter; α95 is radius of cone of 95% confidence.
*S and N indicate location on south and north flank of Arbuckle anticline.
†Ratio of specimens analyzed to specimens used in statistical calculations.
§Semiaxes of oval of 95% confidence about poles.

the diagenetic environment (Turner, 1980). If there was a precursor to the hematite in the Kindblade Formation, the transformation probably occurred relatively soon after dedolomitization, within the resolution of the pole position (~30 Ma).

Several workers have cited strong evidence which suggests that ferroan dolomite is preferentially replaced by calcite under near-surface, oxidizing conditions (e.g., Al-Hashimi and Hemingway, 1973), whereas other authors have documented dedolomitization as occurring in several diagenetic environments (e.g., Land and Prezbindowski, 1981; Budai et al., 1983). The fluids that caused dedolomitization in the Kindblade Formation were probably oxidizing. In addition, the Arbuckle Group contains evaporites in the subsurface, and evidence of replaced evaporites exists in the surface exposures (St. John and Eby, 1978). Dissolution of these evaporites as a result of uplift of the Arbuckle Mountains may have produced calcium-rich solutions, which can cause dedolomitization (e.g., Land and Prezbindowski, 1981).

CONCLUSIONS

Hematite associated with dedolomite in the Kindblade Formation carries a Late Pennsylvanian–Early Permian magnetization. This study represents an example of how one can absolutely date a diagenetic event. The approach requires that a stable magnetization can be isolated, the magnetic mineralogy identified, and the origin of the magnetization determined and, if the magnetization is a CRM, related to a specific diagenetic mineral. Dating also requires a reliable APWP and is enhanced if tests that constrain the timing of the magnetization (i.e., fold tests) can be applied.

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Reviewer's comment

An innovative approach to absolute dating of a diagenetic event.

John King